binds to an actin and activates a contraction – relaxation cycle which is followed by a power stroke.

The power stroke (cross-bridge tilting) begins after the myosin head and hinge region tilt from a 90° angle to a 45° angle. The cross-bridge transforms into strong, high-force bond which allows the myosin head to swivel. When the myosin head swivels it pulls the attached actin filament towards the centre of the A-band. The myosin returns back to its relaxed state and releases ADP and phosphate ion. A new ATP molecule then binds to the head of the myosin and the cross-bridge is broken. At the end of each power stroke, each myosin head detaches from actin, then swivels back and binds to a new actin molecule to start another contraction cycle. This movement is similar to the motion of an oar on a boat. At the end of each power stroke, each myosin head detaches from actin, then swivels back and binds to a new actin molecule to start another contraction cycle. The power stroke repeats many times until a muscle fibre contracts. The myosin heads bind, push and release actin molecules over and over as the thin filaments move toward the centre of the sarcomere. The repeated formation of cross-bridge cycles cause the sliding of the filaments only but there is no change in the lengths of either the thick or thin filaments. The Z- discs attached to the actin filaments are also pulled inwards from both the sides, causing the shortening of the sarcomere (i.e. contraction). This process continues as long as the muscle receives the stimuli and with a steady flow of calcium ions. When motor impulse stops, the calcium ions are pumped back into the sarcoplasmic reticulum which result in the masking of the active sites of the actin filaments. The myosin head fails to bind with the active sites of actin and these changes cause the return of Z-discs back to their original position, i.e. relaxation. (Figure 9.3)

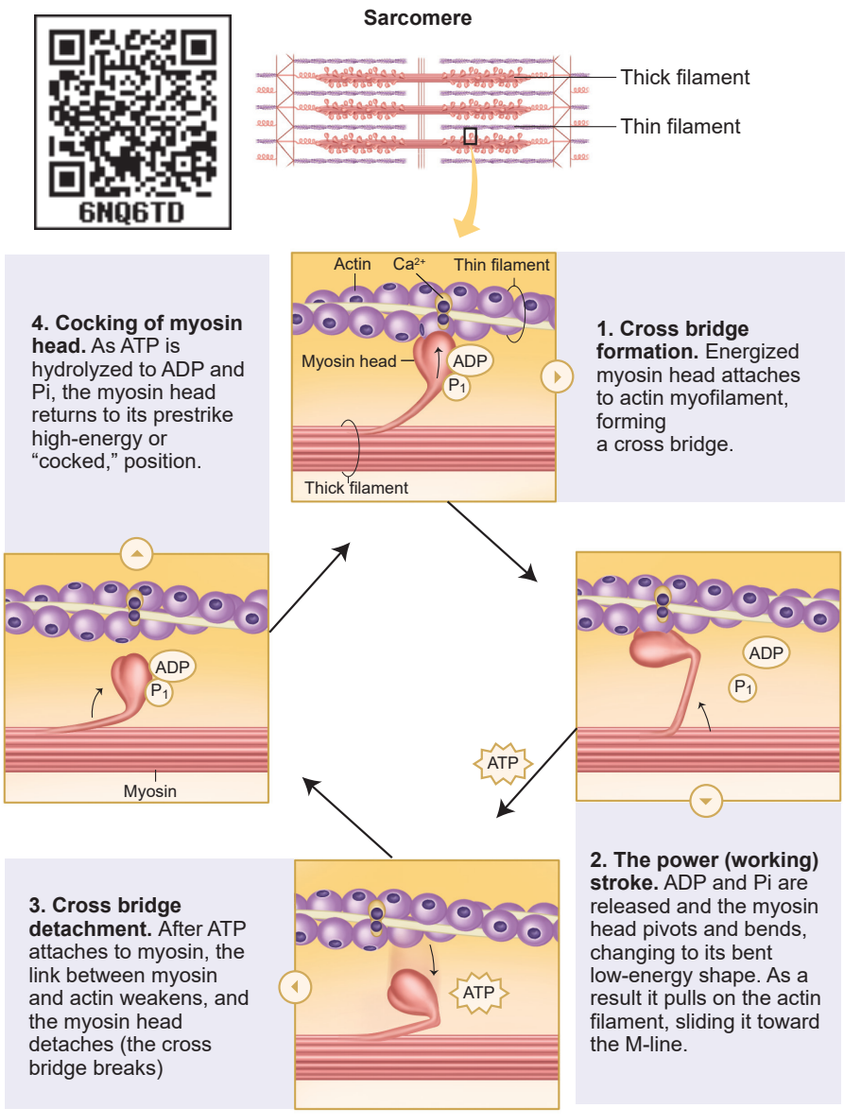


Figure 9.3 Cross-bridge cycle of muscle contraction

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